■ ANIMATRONIC BIRD SHOW - MASTER HARDWARE WIRING SCHEMATIC

Complete Pinout Map for Raspberry Pi 4/5, PCA9685 PWM Driver, Dual L298N Motor Controllers, Relays & Power Rails

■ Core System Hardware Overview

Module / Component	Hardware Role & Signals	Power Requirements
Raspberry Pi 4 / 5	Executes master server, Gemini AI, I2C bus (Pin 3 SDA, Pin 5 SCL), L298N triggers & Relays	+5V @ 3A DC
PCA9685 16-Ch PWM Driver	I2C Address 0x40. Drives 16 PWM channels (3-DOF perches, 2-DOF spotlights, base turntable)	+5V Logic / +5V-+6V 12A V+
Dual L298N Motor Drivers	Driver #1 (Parrot 1) & Driver #2 (Parrot 2) Mouth Solenoids, Blinking Eyes & Flapping Wings	+5V/+6V High-Current Rail
2-Ch Relays & Transistor Driver	Relay #1 (GPIO 17) Left Bird Chirp, Relay #2 (GPIO 27) Right Bird Chirp, Center Bird Gearmotor (GPIO 22)	+5V Logic VCC

■ Complete GPIO & PCA9685 Pinout Lookup Table

Header / Module	Pin / Ch #	Signal Name	Connected Hardware Module	Terminal	Function Description
Raspberry Pi	**Pin 3**	GPIO 2 (SDA1)	PCA9685 PWM Driver	SDA Terminal	I2C Serial Data Bus for 16-Ch Servo Controller
Raspberry Pi	**Pin 5**	GPIO 3 (SCL1)	PCA9685 PWM Driver	SCL Terminal	I2C Serial Clock Bus for 16-Ch Servo Controller
Raspberry Pi	**Pin 11**	GPIO 17	2-Channel Relay Board	Relay #1 IN	Front Left Small Bird Chirp Sound Relay
Raspberry Pi	**Pin 12**	GPIO 18 (PWM)	L298N Driver #1 (P1)	ENA	Parrot 1 Mouth / Eyes Actuator PWM Speed
Raspberry Pi	**Pin 13**	GPIO 27	2-Channel Relay Board	Relay #2 IN	Front Right Small Bird Chirp Sound Relay
Raspberry Pi	**Pin 15**	GPIO 22	Transistor Motor Driver	Base / Gate	Center Small Bird Gearmotor & Beak Driver
Raspberry Pi	**Pin 16**	GPIO 23	L298N Driver #1 (P1)	IN1	Parrot 1 Mouth Open (Normal Polarity)
Raspberry Pi	**Pin 18**	GPIO 24	L298N Driver #1 (P1)	IN2	Parrot 1 Eyes Blink (Reverse Polarity)
Raspberry Pi	**Pin 22**	GPIO 25	L298N Driver #1 (P1)	IN3	Parrot 1 Wing Flap Motor Direction 1
Raspberry Pi	**Pin 24**	GPIO 8	L298N Driver #1 (P1)	IN4	Parrot 1 Wing Flap Motor Direction 2
Raspberry Pi	**Pin 32**	GPIO 12 (PWM)	L298N Driver #2 (P2)	ENA	Parrot 2 Mouth / Eyes Actuator PWM Speed
Raspberry Pi	**Pin 33**	GPIO 13 (PWM)	L298N Driver #2 (P2)	ENB	Parrot 2 Flapping Wings Motor PWM Speed
Raspberry Pi	**Pin 35**	GPIO 19 (PWM)	L298N Driver #1 (P1)	ENB	Parrot 1 Flapping Wings Motor PWM Speed
Raspberry Pi	**Pin 36**	GPIO 16	L298N Driver #2 (P2)	IN1	Parrot 2 Mouth Open (Normal Polarity)
Raspberry Pi	**Pin 37**	GPIO 26	L298N Driver #2 (P2)	IN4	Parrot 2 Wing Flap Motor Direction 2
Raspberry Pi	**Pin 38**	GPIO 20	L298N Driver #2 (P2)	IN2	Parrot 2 Eyes Blink (Reverse Polarity)
Raspberry Pi	**Pin 40**	GPIO 21	L298N Driver #2 (P2)	IN3	Parrot 2 Wing Flap Motor Direction 1
PCA9685 PWM	**Ch 0 - 2**	PWM Servo Out	Parrot 1 (Left) 3-DOF	Servos	Base Turn (Yaw), Bow (Pitch), Shimmy (Roll)
PCA9685 PWM	**Ch 3 - 5**	PWM Servo/LED	Parrot 1 Spotlight	Servos & LED	Spotlight Pan, Tilt & LED Intensity Dimmer

PCA9685 PWM	**Ch 6 - 8**	PWM Servo Out	Parrot 2 (Right) 3-DOF	Servos	Base Turn (Yaw), Bow (Pitch), Shimmy (Roll)
PCA9685 PWM	**Ch 9 - 11**	PWM Servo/LED	Parrot 2 Spotlight	Servos & LED	Spotlight Pan, Tilt & LED Intensity Dimmer
PCA9685 PWM	**Ch 12 - 14**	PWM Servo Out	Center Small Bird 3-DOF	Servos	Base Turn (Yaw), Bow (Pitch), Shimmy (Roll)
PCA9685 PWM	**Ch 15**	PWM Servo Out	Main Stage Turntable	Base Servo	Whole Stage Rotating Turntable Base Servo
Power Rail	**+5V / +6V**	DC Power	PCA9685 V+ & L298N 12V	Terminals	High-Current 12A Servo & Gearmotor Power Rail
Power Rail	**GND Bus**	Common Ground	All Board GND Terminals	GND Bus	Shared Common Reference Ground Bus